

Geometric Control of Quantum and Complex Systems

Habilitation à Diriger des Recherches

Marco Caponigro

8 Novembre 2017
Conservatoire National des Arts et Métiers
Paris

Outline for section 1

1 Geometric control theory

2 Quantum control

- Lie-Galerkin Condition
- Weakly coupled systems

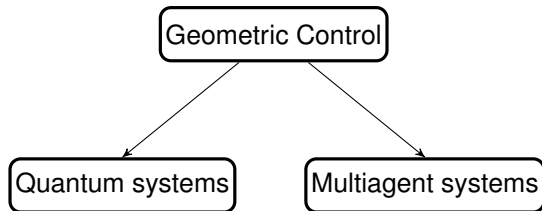
3 Complex systems

- Sparse stabilization and control via leader
- A general result of sparse stabilization
- Mean-Field sparse control

4 Conclusions

Geometric methods in control

- Analysis of the geometric structures
- Study of intrinsic properties
- General results
- Classical and powerful tools for finite dimensional control
- Quite new in the infinite dimensional setting



Outline for section 2

1 Geometric control theory

2 Quantum control

- Lie-Galerkin Condition
- Weakly coupled systems

3 Complex systems

- Sparse stabilization and control via leader
- A general result of sparse stabilization
- Mean-Field sparse control

4 Conclusions

Conservative bilinear systems

Let $\mathcal{H}$ be a complex Hilbert space

$$\frac{d\psi}{dt}(t) = (A + u_1(t)B_1 + \cdots + u_p(t)B_p)\psi(t), \quad (BSE)$$

with initial condition $\psi(0) = \psi_0 \in \mathcal{H}$ with $\|\psi_0\| = 1$ where

- $A, B_1, \dots, B_p$ skew-adjoint operators on $\mathcal{H}$ with domains $D(A), D(B_1), \dots, D(B_p)$ respectively;
- $u = (u_1, \dots, u_p) \in U = [-\delta, \delta]^p$.

Controllability

Given $\psi_0, \psi_1 \in \mathcal{H}$, $\|\psi_0\| = \|\psi_1\| = 1$, find $u : [0, T] \rightarrow U$ such that the solution $\psi(t)$ associated with u with initial data ψ_0 satisfies

$$\psi_1 = \psi(T).$$

Bilinear Schrödinger equation

$$i\frac{d\psi}{dt} = \underbrace{(-\Delta + V)}_{iA} \psi + u \underbrace{W}_{iB} \psi,$$

- $\Omega \subset \mathbb{R}^d$;
- $\psi = \psi(t, x)$ **wave function**, $\psi(t, \cdot) \in L^2(\Omega)$, $\|\psi(t, \cdot)\|_2 = 1$;
- $-\Delta + V$ **Schrödinger operator**;
- $u = u(t) \in U \subset \mathbb{R}$ **control law**;
- $W : \Omega \rightarrow \mathbb{R}$ **controlled potential**.

Bilinear Schrödinger equation

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- $-\Delta + V$ Schrödinger operator;
- $u = u(t) \in U \subset \mathbb{R}$ control law;
- $W : \Omega \rightarrow \mathbb{R}$ controlled potential.

Applications :

- **Photochemistry** (to induce certain chemical reactions with light);
- **Magnetic Resonance** (in order to exploit spontaneous emission);
- **Realization of Quantum Computers** (to stock information).

An obstruction to exact controllability

Let X infinite dimensional Banach space and consider the propagator Υ_t^u of

$$\dot{\psi} = (A + uB)\psi, \quad \psi(0) = \psi_0$$

Theorem (Ball, Marsden, Slemrod, 1982)

Let $A : D(A) \rightarrow X$ be a linear operator generating a C^0 semigroup, and $B : X \rightarrow X$ be **bounded**. Then, for every ψ_0 in X , the attainable set

$$\{\Upsilon_T^u \psi_0 \mid u \in \cup_{r>1} L_{loc}^r, T \geq 0\}.$$

is contained in a countable union of compact subsets of X .

- Idea: the end-point mapping

$$u \mapsto \Upsilon_T^u(\psi_0)$$

is continuous from $\cup_{r>1} L^r$ endowed with the weak-* topology to X .

Continuity of the end-point mapping $\implies$ non-exact controllability

An obstruction to exact controllability

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- Adapted to the Hilbert case by Turinici (2000)
- Recent extension to L^1  L^1

Controllability in infinite dimension: state of the art

- Exact controllability (in $H^3(\Omega)$) for the potential well by analytic methods
 - Beauchard (2005);
 - Beauchard, Coron (2006);
 - Beauchard, Laurent (2010);
 - Morancey (2012),...

Infinite potential well

$$\mathcal{H} = H_{(0)}^3((-1, 1), \mathbb{C})$$

$$i \frac{\partial \psi(x, t)}{\partial t} = \left(-\frac{\partial^2}{\partial x^2} + u(t)x^2 \right) \psi(x, t), \quad x \in (-1, 1), \quad \psi(\pm 1, t) = 0.$$

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 - Beauchard, Coron (2006);
 - Beauchard, Laurent (2010);
 - Morancey (2012),...
- L^2 - and H^s -approximate controllability by Lyapunov methods
 - Ito, Kunisch (2009);
 - Mirrahimi (2009);
 - Nersesyan (2009);
 - Nersesyan, Nersisyan (2010),...
- L^2 -approximate controllability via geometric methods
 - Chambrion, Mason, Sigalotti, Boscain (2009)

Contributions

- Boscain, Caponigro, Chambrion, Sigalotti,
A weak spectral condition for the controllability of the bilinear Schrödinger equation with application to the control of a rotating planar molecule, (CMP 2012, ref **[2]**).
- Boussaïd, Caponigro, Chambrion,
Weakly-coupled systems in quantum control, (IEEE TAC 2013, ref **[4]**).
- Boscain, Caponigro, Sigalotti,
Multi-input Schrödinger equation: controllability, tracking, and application to the quantum angular momentum, (JDEQ 2014, ref **[3]**).
- Caponigro, Sigalotti,
Exact controllability in projections of the bilinear Schrödinger equation, (preprint 2017, ref **[21]**).

Conservative bilinear equation

Let $\mathcal{H}$ be a complex Hilbert space

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with initial condition $\psi(0) = \psi_0 \in \mathcal{H}$ with $\|\psi_0\| = 1$ where

- $A, B_1, \dots, B_p$ skew-adjoint operators on $\mathcal{H}$ with domains $D(A), D(B_1), \dots, D(B_p)$ respectively;
- $u = (u_1, \dots, u_p) \in U = [-\delta, \delta]^p$.

Assumptions

- A has discrete spectrum $(i\lambda_k)_{k \in \mathbb{N}}$;
- $\mathcal{H}$ has an Hilbert basis $\Phi = (\phi_k)_{k \in \mathbb{N}}$ made of eigenfunctions of A ;
- $\phi_k \in D(B_j)$ for every $k \in \mathbb{N}, j = 1, \dots, p$;
- $A + u_1B_1 + \cdots + u_pB_p : \text{span}\{\phi_k \mid k \in \mathbb{N}\} \rightarrow \mathcal{H}$ is essentially skew-adjoint for every $u \in U$.

Solutions and classes of admissible controls

Piecewise constant controls

Every piecewise constant control is admissible (*i.e.* the solution is well-posed).

$$\Upsilon_t^u \psi_0 = e^{(t - \sum_{l=1}^{j-1} t_l)(A + u_1^{(j)} B_1 + \dots + u_p^{(j)} B_p)} \circ \dots \circ e^{t_1(A + u_1^{(1)} B_1 + \dots + u_p^{(1)} B_p)} \psi_0,$$

is the **solution** of (BSE) with initial data $\psi_0 \in \mathcal{H}$ associated with u . Υ_t^u is the **propagator** of (BSE) associated with u .

If $B_1, \dots, B_p$ are A -bounded, see [Definition](#), then

- smooth controls are admissible
- BV controls are admissible $\rightarrow$ mild solutions, see [Details](#)

Approximate controllability

Definition

Let $\mathcal{U} \subset L^\infty$ be a class of admissible controls.

(BSE) is *approximately controllable* by means of controls in $\mathcal{U}$ if for every $\psi_0, \psi_1 \in \mathcal{H}$, with $\|\psi_0\| = \|\psi_1\| = 1$ and every $\varepsilon > 0$ there exist $u \in \mathcal{U}$ and $T > 0$ such that

$$\|\psi_1 - \Upsilon_T^u(\psi_0)\| < \varepsilon.$$

Galerkin approximation

$\Phi = (\phi_k)_{k \in \mathbb{N}}$ is an Hilbert basis of $\mathcal{H}$ made of eigenfunctions of A . Let

$$\mathcal{L}_n = \text{span}\{\phi_1, \dots, \phi_n\}.$$

The **Galerkin approximation** of order n of (BSE) is the system

$$\dot{x} = (A^{(n)} + u_1 B_1^{(n)} + \dots + u_p B_p^{(n)})x, \quad x \in \mathcal{L}_n$$

where

$$A^{(n)} = A|_{\mathcal{L}_n}, B_1^{(n)} = B_1|_{\mathcal{L}_n}, \dots, B_p^{(n)} = B_p|_{\mathcal{L}_n}.$$

- Controllability for the Galerkin approximation does not imply controllability for the infinite dimensional system.

Counterexample: the quantum harmonic oscillator

Quantum Harmonic oscillator

$$\mathcal{H} = L^2(\mathbb{R}, \mathbb{C})$$

$$i \frac{\partial \psi(x, t)}{\partial t} = \left(-\frac{\partial^2}{\partial x^2} + x^2 + u(t)x \right) \psi(x, t), \quad x \in \mathbb{R},$$

- Every Galerkin approximation is controllable
- The infinite dimensional system is not controllable (Mirrahimi, Rouchon (2004))

The Lie-Galerkin condition (case $p = 1$)

We consider

$$\Sigma_n = \{|\lambda_j - \lambda_k| \mid j, k = 1, \dots, n\}.$$

And for every $\sigma \in \Sigma_n$, every $m \in \mathbb{N}$, and every $m \times m$ matrix M , let [► Motivation](#)

$$\mathcal{E}_\sigma(M) = (M_{j,k} \delta_{\sigma, |\lambda_j - \lambda_k|})_{j,k=1}^m.$$

Define

$$\Xi_n = \left\{ \sigma \in \Sigma_n \mid \right. \\ \left. \exists M \in \mathfrak{u}(n) \text{ s.t. } \mathcal{E}_\sigma(B^{(N)}) = \left(\begin{array}{c|c} M & 0 \\ \hline 0 & * \end{array} \right) \text{ for every } N > n \right\}.$$

Let

$$\mathcal{V}_n^0 = \left\{ A^{(n)} \right\} \cup \left\{ \mathcal{E}_\sigma(B^{(n)}) \mid \sigma \in \Xi_n, \sigma \neq 0 \right\}.$$

The Lie-Galerkin condition (case $p = 1$)

Let

$$\mathcal{V}_n^0 = \left\{ A^{(n)} \right\} \cup \left\{ \mathcal{E}_\sigma(B^{(n)}) \mid \sigma \in \Xi_n, \sigma \neq 0 \right\}.$$

Definition

The Lie-Galerkin condition holds if for every $n_0 \in \mathbb{N}$ there exists $n > n_0$ such that

$$\text{Lie}\mathcal{V}_n^0 \supseteq \mathfrak{su}(n).$$

Example: existence of a nonresonant chain of connectedness (case $p = 1$)

(A, B) admits a **connectedness chain** $S \subset \mathbb{N}^2$ if B (in the basis of eigenfunctions of A) is “connected”. [► Formal definition](#)

A **connectedness chain** for (A, B) , S is said to be **nonresonant** if

$$|\lambda_j - \lambda_k| \neq |\lambda_\ell - \lambda_m|$$

for every $(j, k) \in S$, $(\ell, m) \in \mathbb{N}^2$, $\{j, k\} \neq \{\ell, m\}$.

Theorem (Boscain, C., Chambrion, Sigalotti, 2012)

If $\langle \phi_j, B\phi_k \rangle = 0$ whenever $j \neq k$ and $\lambda_j = \lambda_k$ (i.e. no internal coupling) and if there exists a nonresonant connectedness chain then the Lie-Galerkin condition holds.

Example: The bipolar molecule

$$i \frac{\partial \psi(\theta, \varphi, t)}{\partial t} = -\Delta \psi(\theta, \varphi, t) + (u_1(t) \sin \theta \cos \varphi + u_2(t) \sin \theta \sin \varphi + u_3(t) \cos \theta) \psi(\theta, \varphi, t),$$

where θ, φ are the spherical coordinates and $\psi \in L^2(S^2, \mathbb{C})$

Δ is the Laplace–Beltrami operator on the sphere S^2

$$\Delta = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2},$$

- the ℓ -th eigenvalue is $-i\ell(\ell + 1)$
- the ℓ -th eigenfunctions associated to $-i\ell(\ell + 1)$ are the spherical harmonics $Y_\ell^m(\theta, \varphi)$ for $m = -\ell, \dots, \ell$,
- the ℓ -th eigenvalue has multiplicity $2\ell + 1$.

Approximate controllability

Theorem (Boscain, C., Sigalotti, 2014)

If the Lie-Galerkin condition holds then the system is approximately controllable.

Approximate controllability

Theorem (Boscain, C., Sigalotti, 2014)

If the Lie-Galerkin condition holds then the system is approximately controllable.

- General result;
- Generic condition;
- Estimates on the L^1 norm of the control;
- Approximate simultaneous controllability holds.

Approximate simultaneous controllability

Given $\varepsilon > 0$, $\psi^1, \dots, \psi^m \in \mathcal{H}$, $\hat{\Upsilon} \in \mathbf{U}(\mathcal{H})$ find $u : [0, T] \rightarrow U$ such that

$$\|\hat{\Upsilon}(\psi^j) - \Upsilon_T^u(\psi^j)\| < \varepsilon \quad j = 1, \dots, m.$$

Theorem (C., Sigalotti, 2017)

If smooth controls are admissible and if Lie-Galerkin condition holds then the system is approximately controllable by means of smooth controls.

Exact controllability in projections

Continuity assumption (CA)

$$u_n \xrightarrow{n \rightarrow \infty} u \text{ in } L^2([0, T]) \implies \Upsilon_t^{u_n} \phi \xrightarrow{n \rightarrow \infty} \Upsilon_t^u \phi,$$

for every $\phi \in \mathcal{H}$ and uniformly with respect to $t \in [0, T]$.

Theorem (C., Sigalotti, 2017)

Assume that the Lie-Galerkin condition holds and that the end-point mapping is continuous as in (CA). Then for every

- $n \in \mathbb{N}$,
- $\varepsilon > 0$,
- initial condition $\psi_0 \in \mathcal{H}$ with $\|\psi_0\| = 1$,
- final condition $\psi_1 \in \mathcal{H}$ such that $\|\psi_1\| = 1$ and $\|\Pi_n(\psi_1)\| < 1$,

there exists $u : [0, T] \rightarrow U$, $u \in \mathcal{U}$ such that

$$\Pi_n(\Upsilon_T^u(\psi_0)) = \Pi_n(\psi_1) \quad \text{and} \quad \|\Upsilon_T^u(\psi_0) - \psi_1\| < \varepsilon.$$

- (CA) $\implies$ non-exact controllability

Weakly coupled systems

- $i(A + u_1 B_1 + \cdots + u_p B_p)$ is bounded from below for every $u \in U$
- λ_j is nondecreasing and unbounded

Definition (k -weakly coupled)

The system (A, B) is k -weakly coupled if

- $D(|A + u_1 B_1 + \cdots + u_p B_p|^{k/2}) = D(|A|^{k/2})$ for every $u \in \mathbb{R}^p$
- there exists C such that

$$|\Re \langle |A|^k \psi, B_j \psi \rangle| \leq C |\langle |A|^k \psi, \psi \rangle|, \quad \forall \psi \in D(|A|^k), j = 1, \dots, p.$$

Examples:

- B is relatively bounded wrt A .
- $iA = -\Delta + V$, $iB = W$ and $V, W \in C^{2k}(\Omega)$, Ω compact.

Growth of the $|A|^{k/2}$ -norm

$$\|\psi\|_{k/2} = \| |A|^{k/2} \psi \|^2 = |\langle |A|^k \psi, \psi \rangle| = \sum_{n \in \mathbb{N}} \lambda_n^k |\langle \phi_k, \psi \rangle|$$

Proposition (Boussaïd, C, Chambrion, 2013)

Let $(A, B_1, \dots, B_p, U, \Phi)$ be k -weakly coupled. Then, for every

- $\psi_0 \in D(|A|^{k/2})$;
- $K > 0$;
- $T \geq 0$;
- $u = (u_1, \dots, u_p)$ piecewise constant such that $\sum_{l=1}^p \|u_l\|_{L^1} \leq K$,

one has

$$\|\Upsilon_T^u(\psi_0)\|_{k/2} \leq e^{cK} \|\psi_0\|_{k/2},$$

c not depending on u and T .

The regularity of the systems is an obstacle to the exact controllability

Good Galerkin approximation

Let $\mathcal{L}_N = \text{span}\{\phi_1, \dots, \phi_N\}$ and $X_u^{(N)}$ be the propagator of

$$\dot{x} = (A|_{\mathcal{L}_N} + u_1 B_1|_{\mathcal{L}_N} + \dots + u_p B_p|_{\mathcal{L}_N})x, \quad x \in \mathcal{L}_N.$$

Theorem (Boussaïd, C, Chambrion, 2013)

Let $(A, B_1, \dots, B_p, U, \Phi)$ be k -weakly coupled and let $B_1, \dots, B_p$ be $|A|^s$ -bounded, $s < k$. Then, for every

- $\varepsilon > 0$;
- $\psi_0 \in D(|A|^{k/2})$;
- $K > 0$;
- $T \geq 0$;

there exists $N = N(\varepsilon, K, \psi_0)$ such that

$$\sum \|u_j\|_{L^1} \leq K \implies \|\Upsilon_t^u(\psi_0) - X_u^{(N)}(t)\psi_0\|_s < \varepsilon, \quad t \geq 0.$$

Good Galerkin approximation

$$\sum \|u_j\|_{L^1} \leq K \implies \|\Upsilon_t^u(\psi_0) - X_u^{(N)}(t)\psi_0\|_s < \varepsilon, \quad t \geq 0.$$

- A priori estimates in numerical and physical simulations.
- Convergence of controllability strategies:
 - A Bang-Bang theorem for weakly coupled systems
 - Approximate controllability in stronger norms

Open problems

- Bilinear Schrödinger Equation
 - Time estimates (from above and from below)
 - L^1 -optimal control
 - Continuity of the end-point mapping
- Other PDEs
 - Hypoelliptic Schrödinger Equation
 - Open systems (Lindblad equations)
 - Wave equations (bilinear and nonlinear)

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Multi-agent systems

System of N ODEs. The dynamics of the component $x_i \in \mathbb{R}^d$ are influenced by the state of the other $N - 1$ components

$$\dot{x}_i = \frac{1}{N} \sum_{j \neq i} F(x_i, x_j), \quad i = 1, \dots, N,$$

- x_i is the state of agent i ;
- interaction rule $F \in \text{Lip}(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$;
- stable configurations are sometimes called *consensus*.

Applications

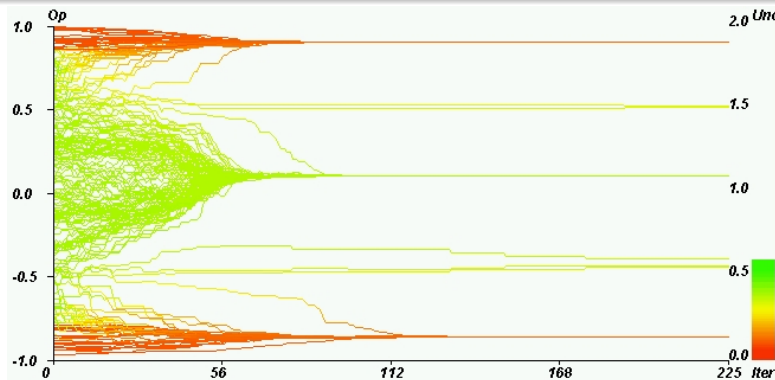
- Robotics: *e.g.* robot/drone swarming;
- Aerospace: *e.g.* satellite coordination;
- Social sciences: *e.g.* opinion formation;
- Finance: *e.g.* market evolution.

First order models

- Also called opinion formation models (hence the term *consensus*);
- Aggregation or separation emerge.

Hegselmann, Krause (2002)

$$\dot{x}_i = \sum_{|x_i - x_j| < 1} a_{ij}(x_j - x_i) \quad \text{for } i = 1, \dots, N$$



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- in $\mathbb{R}^N$: Krause (2000), Hegselmann, Krause (2002), Blondel, Hendricks, Tsitsiklis (2010), ecc...
- in Riemannian manifolds : C., Lai, Piccoli (DCDS 2015, ref **[17]**)

Second order models

$$\ddot{x}_i = \sum_j f(x_i, x_j, \dot{x}_i, \dot{x}_j), \quad i = 1, \dots, N,$$

- Consensus is usually achieved on the $\dot{x}$ variable
- Alignment, separation, cohesion, friction, self-propulsion, etc...

Cucker, Smale (2007)

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = \frac{1}{N} \sum_{j=1}^N a(\|x_i - x_j\|)(v_j - v_i) \end{cases} \quad (CS)$$

- Reynolds (1987), Vicsek et al. (1995), D'Orsogna et al. (2006), Cucker, Smale (2007), Cucker, Dong (2011), Motsch, Tadmor (2011), Haskovec (2013), Ahn et al. (2014)...

Modelling and control

- Modelling is an open field of research. Main challenges are
 - in finite dimension: high dimensionality, noise, “new” phenomena (nudging, black swans), etc...
 - in infinite dimension: mean-field limits, kinetic equations, etc...

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 - in finite dimension: high dimensionality, noise, “new” phenomena (nudging, black swans), etc...
 - in infinite dimension: mean-field limits, kinetic equations, etc...
- External control is a rather new subject
 - classical tools apply, see [Jurdjevic-Quinn](#) ;
 - new challenges coming from optimal control.

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Sparse controllability problem

Is it possible to control the system with a control acting on one component only ?

Contributions

- Cucker–Smale model: sparse stabilization, optimal control (MCRF 2013, ref **[15]**)
- General alignment models: sparse stabilization, optimal control, sparse controllability (M3AS 2015, ref **[14]**)

$$\begin{cases} \dot{x}_i(t) = v_i(t), \\ \dot{v}_i(t) = \frac{1}{N} \sum_{j=1}^N a_{ij}(v_j(t) - v_i(t)) + u_i, \end{cases}$$

Contributions

- Hegselmann–Krause model: stabilization via leader, optimal control via leader (MCRF 2015, ref **[23]**)
- Heider balance model: control via leader (EPJ 2015, ref **[22]**)

$$\begin{cases} \dot{x}_0(t) = u(t), \\ \dot{x}_i(t) = \frac{1}{N} \sum_{j=1}^N a_{ij}(x_j(t) - x_i(t)) + c_1(x_0(t) - x_i(t)), \end{cases}$$

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A general result of sparse stabilization

Consider

- $\mathbb{U} \subset \mathbb{R}^m$ compact such that $0 \in \mathring{\mathbb{U}}$
- $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $f(0) = 0$
- the control affine system on $\mathbb{R}^n$

$$\dot{x} = f(x) + \sum_{i=1}^m u_i g_i(x), \quad u \in \mathbb{U}$$

Assume that there exists $V : \mathbb{R}^n \rightarrow \mathbb{R}$ smooth such that

- i) **positive:** $V(0) = 0$ and $V(x) > 0$ if $x \neq 0$
- ii) **proper:** $V(x) \rightarrow \infty$ as $\|x\| \rightarrow \infty$
- iii) **dissipative:** $L_f V(x) \leq 0$ for every $x \in \mathbb{R}^n$

where

$$L_f V(x) = \left. \frac{d}{dt} V(e^{tf} x) \right|_{t=0}$$

iv) (Weak) Jurdjevic–Quinn condition

$$\{x \in \mathbb{R}^n \mid L_f V(x) = 0 = L_{g_i} V(x), \forall i = 1, \dots, m\} = \{0\}$$

Theorem (Jurdjevic–Quinn)

The system $\dot{x} = f(x) + \sum_{i=1}^m u_i g_i(x)$ is globally asymptotically stabilizable by means of the control defined by

$$u_i(x) = -L_{g_i} V(x), \quad i = 1, \dots, m$$

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- Based on LaSalle Invariance principle

$$\frac{d}{dt} V(x(t)) = L_f V(x(t)) - \sum_{i=1}^m |L_{g_i} V(x(t))|^2 \leq 0$$

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The system $\dot{x} = f(x) + \sum_{i=1}^m u_i g_i(x)$ is globally asymptotically stabilizable by means of the control defined by

$$u_i(x) = -L_{g_i} V(x), \quad i = 1, \dots, m$$

Can we stabilize by means of a **sparse** control?

Periodic time-varying

Fix a sampling time $\tau > 0$. Consider the time-varying feedback $u(t, x) = (u_1(t, x), \dots, u_m(t, x))$ defined by

$$u_i(t, x) = \begin{cases} -L_{g_i} V(x(t)), & \text{if } t \in [(km + i - 1)\tau, (km + i)\tau) \\ 0 & \text{otherwise.} \end{cases}$$

for $k \in \mathbb{N}$ and $i = 1, \dots, m$

Theorem (C., Piccoli, Rossi, Trélat, 2017)

If 0 is locally attractive then for every x_0 there exists $\tau > 0$ such that $u(t, x)$ asymptotically stabilizes the system to 0.

- Averaging method
- Local attractivity is necessary

Steepest descent feedback: sampling and holding

Consider the feedback $u(x) = (u_1(x), \dots, u_m(x))$ defined by

$$u_i(x) = -L_{g_i} V(x) \quad \text{and} \quad u_j = 0 \quad \text{for } j \neq i,$$

where $i \in \{1, \dots, m\}$ is the smallest integer such that

$$|L_{g_i} V(x)| \geq |L_{g_j} V(x)|, \quad \forall j \neq i.$$

for $k \in \mathbb{N}$ and $i = 1, \dots, m$

Theorem (C., Piccoli, Rossi, Trélat, 2017)

If 0 is locally attractive then for every x_0 there exists $\tau > 0$ such that the sampling solution associated with $u(x)$ asymptotically tends to 0.

- Steepest descent feedbacks are stabilizers
- Local attractivity is not necessary but stabilization is in an approximate sense (i.e. in the sample-and-hold sense).

Steepest descent feedback: hysteresis

- Fix $\varepsilon \in (0, 1)$
- Control, switchings $(t_n)_{n \in \mathbb{N}}$, and solution constructed iteratively on $[t_n, t_{n+1})$.
- Let $i = 1, \dots, m$ be the smallest s.t.

$$|L_{g_i} V(x(t_n))| \geq |L_{g_j} V(x(t_n))|, \quad \text{for every } j \neq i.$$

- Define $u_i(x) = -L_{g_i} V(x)$ and $u_j = 0$ for $j \neq i$,
- Switch as soon as

$$|L_{g_i} V(y(t))| \leq (1 - \varepsilon) |L_{g_j} V(y(t))|, \quad \text{for some } j \neq i.$$

Theorem (C., Piccoli, Rossi, Trélat, 2017)

For every $\varepsilon \in (0, 1)$ and $x_0 \in \mathbb{R}^n$, u asymptotically stabilizes the system to 0.

- The three results apply to first-order and second-order models, see [Example 1](#) and [Example 2](#)

Mean-Field sparse control

Control of a transport PDE with non-local term

$$\partial_t \mu + \nabla \cdot ((f[\mu] + ug[\mu])\mu) = 0, \quad \mu(0) = \mu_0,$$

- μ probability measure on $\mathbb{R}^d$ with compact support (i.e. $\mu \in \mathcal{P}_c(\mathbb{R}^d)$)
- $f, g : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \text{Lip}(\mathbb{R}^d, \mathbb{R}^d)$
- the control $u(t, x) = \chi_{\omega(t)}(x)v(t, x)$ satisfies the constraint
 - $|\omega(t)| \leq c,$
 - $\|u(t, \cdot)\|_{L^\infty} \leq 1.$

In

- Caponigro, Piccoli, Rossi, Trélat, *Mean-field sparse Jurdjevic–Quinn control*, (M3AS 2017, ref [18]).

Open problems

- Distributed sparse control
- Optimal control of several leaders in competition/cooperation → Differential games
- Multiscale modeling, coupling ODEs and PDEs → Crowd dynamics
- Sparse control of multiscale dissipative systems, etc...

Outline for section 4

1 Geometric control theory

2 Quantum control

- Lie-Galerkin Condition
- Weakly coupled systems

3 Complex systems

- Sparse stabilization and control via leader
- A general result of sparse stabilization
- Mean-Field sparse control

4 Conclusions

Geometric control



BOSCAIN



SIGALOTTI



CHAMBRION



BOUSSAÏD

Quantum



GHEZZI



TRÉLAT



BORZI



WONGKAEW



PICCOLI

Complex

For every u in $L^1_{loc}([0, +\infty), \mathbb{R})$, p in $\mathbb{N}$, $t \geq 0$, and ψ_0 in X

$$\Upsilon_{t,0}^u \psi_0 = \sum_{p=0}^{\infty} W_p(t, u) \psi_0.$$

Where $W_p(t, u) : X \rightarrow X$ is defined recursively, for ψ in X , by

$$\begin{aligned} W_0(t, u) \psi &= e^{(t-s)A} \psi \\ W_p(t, u) \psi &= \int_0^t e^{(t-s)A} B W_{p-1}(s, u) \psi u(s) ds, \quad \text{for } p \geq 1, \end{aligned}$$

For every u in $L^1_{loc}([0, +\infty), \mathbb{R})$, p in $\mathbb{N}$, $t \geq 0$, and ψ_0 in X

$$\Upsilon^u_{t,0}\psi_0 = \sum_{p=0}^{\infty} W_p(t, u)\psi_0.$$

For every u in $L^1_{loc}([0, +\infty), \mathbb{R})$, $p \in \mathbb{N}$, and $t \geq 0$

$$\|W_p(t, u)\| \leq \frac{Me^{\omega t} \|B\|^p (\int_0^t |u(s)| ds)^p}{p!}.$$

Lemma

For every p in $\mathbb{N}$, $T \geq 0$ and $K \geq 0$, and ψ in X the set

$$\{W_p(t, u)\psi \mid 0 \leq t \leq T, \|u\|_{L^1} \leq K\},$$

is totally bounded

Averaging: idea for $p = 1$

- **Step 1:** $y(t) = e^{-tA}\psi(t)$ is solution of

$$\dot{y}(t) = \Theta(t, u(t))y(t).$$

where

$$\langle \phi_k, \Theta(t, u)\phi_j \rangle = e^{i(\lambda_k - \lambda_j)t} u B_{jk}$$

- **Step 1:** For every σ take $v^N(t)$ such that

- $v^N(t)$ is $2\pi/\sigma$ -periodic
- $\int_0^{2\pi/\sigma} v^N(t) e^{i\sigma t} dt = 1$;
- $\int_0^{2\pi/\sigma} v^N(t) e^{im\sigma t} dt = 0$ for every $m \neq 1$ such that $m\sigma \in \Sigma_N$.

Then, for every $\sigma' \neq \sigma$,

$$\frac{1}{K} \int_0^{Kt} v^N(s) e^{i\sigma' s} ds \rightarrow 0 \text{ as } K \rightarrow \infty.$$

Connectedness chain

$S \subset \mathbb{N}^2$ is a **connectedness chain** for (A, B) if

- $\langle \phi_\alpha, B\phi_\beta \rangle \neq 0$ for every $(\alpha, \beta) \in S$;
- for every $j \leq k \in \mathbb{N}$, there exist $(\alpha_1, \beta_1), \dots, (\alpha_p, \beta_p)$ in S such that

$$j = \alpha_1, \quad \beta_1 = \alpha_2 \quad \dots \quad \beta_{p-1} = \alpha_p, \quad \beta_p = k.$$

Relative boundedness

Definition

B is said *relatively bounded with respect to A* , or *A -bounded*, if $D(A) \subset D(B)$ and there exist $a, b > 0$ such that for every ψ in $D(A)$,

$$\|B\psi\| \leq a\|A\psi\| + b\|\psi\|. \quad (1)$$

The A -bound of B is the smallest constant a the inequality holds.

If A is skew-adjoint, B skew-symmetric, and B is relatively bounded with respect to A , then

$A + uB$ is skew-adjoint with domain $D(A)$ for $|u| < 1/a$.

Consequence of Kato–Rellich Theorem

Solutions and classes of admissible controls II

Smooth controls

If $B_1, \dots, B_p$ are A -bounded with A -bound smaller than $1/\delta$. Then given $T > 0$, every $u \in C^1([0, T], U)$ is admissible.

If, moreover, $D(A) = D(A + u_1 B_1 + \dots + u_p B_p)$ for every $u \in U$ then, given $\psi_0 \in D(A)$, $\Upsilon_t^u(\psi_0)$ satisfies (BSE) for every $t \in [0, T]$.

BV controls

If $B_1, \dots, B_p$ are A -bounded with A -bound smaller than $1/\delta$. Assume moreover that $D(A) = D(A + u_1 B_1 + \dots + u_p B_p)$ for every $u \in U$. Then, given $T > 0$, every $u \in BV([0, T], U)$ is admissible and the associated solution satisfies

$$\psi(t) = e^{tA}\psi_0 + \int_0^t e^{(t-s)A}(u_1(s)B_1 + \dots + u_p(s)B_p)\psi(s)ds,$$

for every $t \in [0, T]$.

- BV controls $\rightarrow$ *Mild* solutions.

Example: Opinion formation models

Hegselmann, Krause (2002)

$$\dot{x}_i = \sum_{|x_i - x_j| < 1} a_{ij}(x_j - x_i) + u_i \quad \text{for } i = 1, \dots, N$$

for $a_{ij} \geq 0$.

The Lyapunov function is

$$V(t) := \sum_{i,j=1}^N \|x_i(t) - x_j(t)\|^2$$

Example: Alignment models

Motsch–Tadmor (2011)

$$\begin{cases} \dot{x}_i(t) = v_i(t), \\ \dot{v}_i(t) = \frac{1}{N} \sum_{j=1}^N a_{ij}(v_j(t) - v_i(t)) + u_i, \end{cases}$$

for $i = 1, \dots, N$, where

$$a_{ij} = \frac{a(\|x_j(t) - x_i(t)\|)}{\sum_k a(\|x_k(t) - x_i(t)\|)}$$

for $a \in C^1((0, +\infty))$ positive nonincreasing .

The Lyapunov function is

$$V(t) = \frac{1}{2N^2} \sum_{i,j=1}^N \|v_i(t) - v_j(t)\|^2$$

Bilinear Schrödinger Equation

$$\frac{d\psi}{dt}(t) = (A + u(t)B)\psi(t), \quad u(t) \in U.$$

$$y(t) = e^{-tA} \Upsilon_t^u(y(0))$$

Bilinear Schrödinger Equation

$$\frac{d\psi}{dt}(t) = (A + u(t)B)\psi(t), \quad u(t) \in U.$$

Interaction Framework

$$\dot{y}(t) = u(t)\Theta(t)y(t),$$

where

$$\langle \phi_k, \Theta(t)\phi_j \rangle = e^{i(\lambda_k - \lambda_j)t} B_{jk}.$$

$$y(t) = e^{-tA} \Upsilon_t^u(y(0))$$

Bilinear Schrödinger Equation

$$\frac{d\psi}{dt}(t) = (A + u(t)B)\psi(t), \quad u(t) \in U.$$

Interaction Framework

$$\dot{y}(t) = u(t)\Theta(t)y(t),$$

where

$$\langle \phi_k, \Theta(t)\phi_j \rangle = e^{i(\lambda_k - \lambda_j)t} B_{jk}.$$

$$N < \infty$$

Galerkin approximation

$$\dot{y}(t) = u(t)\Theta^N(t)y(t).$$

$$y(t) = e^{-tA} \Upsilon_t^u(y(0))$$

Bilinear Schrödinger Equation

$$\frac{d\psi}{dt}(t) = (A + u(t)B)\psi(t), \quad u(t) \in U.$$

Interaction Framework

$$\dot{y}(t) = u(t)\Theta(t)y(t),$$

where

$$\langle \phi_k, \Theta(t)\phi_j \rangle = e^{i(\lambda_k - \lambda_j)t} B_{jk}.$$

$$N < \infty$$

Galerkin approximation

$$\dot{y}(t) = u(t)\Theta^N(t)y(t).$$

System on $SU(n)$

$$\dot{x} = M(t)x, \quad M(t) \in \mathcal{V}_n,$$

where

$$\mathcal{V}_n = \{A^{(n)}\} \cup \{\mathcal{E}_\sigma(B^{(n)}) \mid \sigma \in \Xi_n, \sigma \neq 0\}.$$

Averaging

$$y(t) = e^{-tA} \Upsilon_t^u(y(0))$$

Bilinear Schrödinger Equation

$$\frac{d\psi}{dt}(t) = (A + u(t)B)\psi(t), \quad u(t) \in U.$$

Interaction Framework

$$\dot{y}(t) = u(t)\Theta(t)y(t),$$

where

$$\langle \phi_k, \Theta(t)\phi_j \rangle = e^{i(\lambda_k - \lambda_j)t} B_{jk}.$$

$$N < \infty$$

definition of Ξ_n

System on $SU(n)$

$$\dot{x} = M(t)x, \quad M(t) \in \mathcal{V}_n,$$

where

$$\mathcal{V}_n = \{A^{(n)}\} \cup \{\mathcal{E}_\sigma(B^{(n)}) \mid \sigma \in \Xi_n, \sigma \neq 0\}.$$

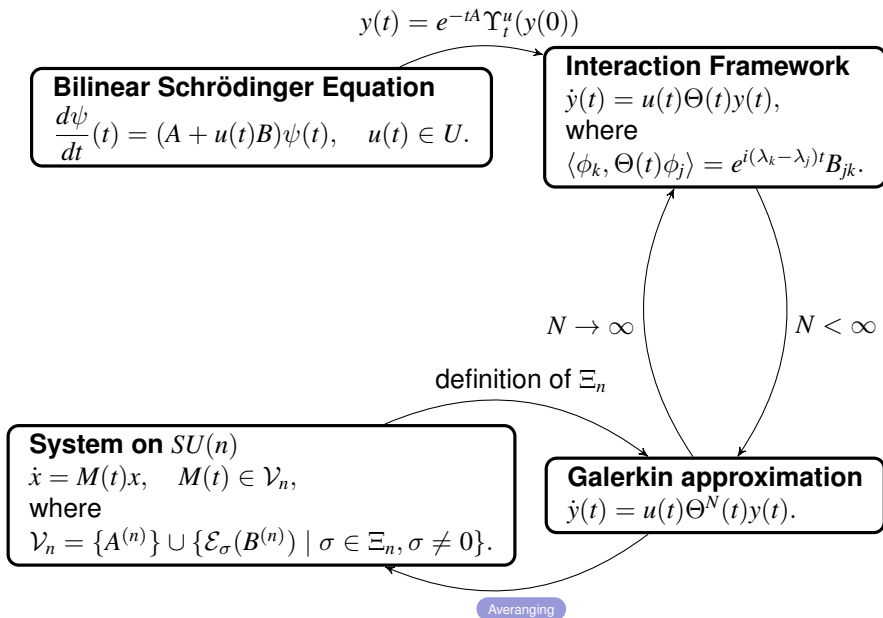
Galerkin approximation

$$\dot{y}(t) = u(t)\Theta^N(t)y(t).$$

Averaging

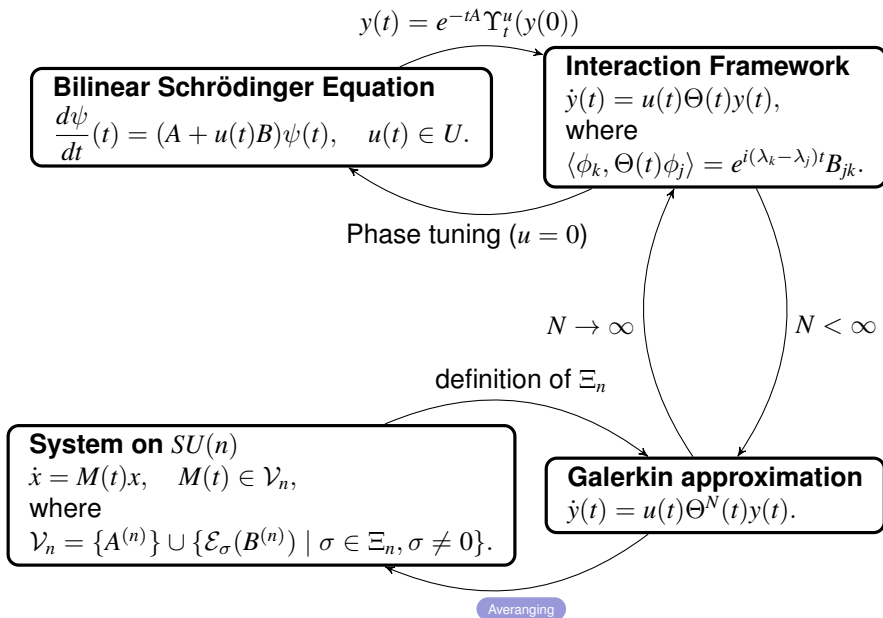
The Lie-Galerkin method

◀ Back



The Lie-Galerkin method

◀ Back



multi-agent models

N interacting agents. The dynamics of the state $x_i \in \mathbb{R}^d$ are influenced by the state of the other $N - 1$ agents,

$$\dot{x}_i = \frac{1}{N} \sum_{j \neq i} F(x_i, x_j) \quad i = 1, \dots, N,$$

- interaction rule $F \in \text{Lip}(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$;

kinetic multi-agent models

N interacting agents. The dynamics of the state $x_i \in \mathbb{R}^d$ are influenced by the state of the other $N - 1$ agents,

$$\dot{x}_i = \frac{1}{N} \sum_{j \neq i} F(x_i, x_j) \quad i = 1, \dots, N,$$

- interaction rule $F \in \text{Lip}(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$;

$$\partial_t \mu + \nabla \cdot ((f[\mu] \quad)\mu) = 0.$$

- μ probability measure on $\mathbb{R}^d$ with compact support ($\mu \in \mathcal{P}_c(\mathbb{R}^d)$)
- $f[\mu](x) = \int F(x, y) d\mu(y)$

Controlled kinetic multi-agent models

N interacting agents. The dynamics of the state $x_i \in \mathbb{R}^d$ are influenced by the state of the other $N - 1$ agents,

$$\dot{x}_i = \frac{1}{N} \sum_{j \neq i} F(x_i, x_j) + u_i \frac{1}{N} \sum_{j \neq i} G(x_i, x_j), \quad i = 1, \dots, N,$$

- interaction rule $F \in \text{Lip}(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$;
- control potential $G \in \text{Lip}(\mathbb{R}^d \times \mathbb{R}^d, \mathbb{R}^d)$;
- controls $(u_1, \dots, u_N) \in U \subset (\mathbb{R}^d)^N$

$$\partial_t \mu + \nabla \cdot ((f[\mu] + ug[\mu])\mu) = 0.$$

- μ probability measure on $\mathbb{R}^d$ with compact support ($\mu \in \mathcal{P}_c(\mathbb{R}^d)$)
- $f[\mu](x) = \int F(x, y) d\mu(y)$
- $g[\mu](x) = \int G(x, y) d\mu(y)$

Assumptions

$$\partial_t \mu + \nabla \cdot ((f[\mu] + ug[\mu])\mu) = 0.$$

(H1) $f, g : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \text{Lip}(\mathbb{R}^d, \mathbb{R}^d)$

(H2) there exists $R > 0$ such that if $\mu \in \mathcal{P}_c(\overline{B(0, R)})$

$$f[\mu](x) = g[\mu](x) = 0 \quad \forall x \notin \overline{B(0, R)}$$

(H3) there exist $L > 0$ and $p \geq 1$ such that, for all $\mu, \nu \in \mathcal{P}_c(\overline{B(0, R)})$ and $x \in \mathbb{R}^d$,

$$|f[\mu](x) - f[\nu](x)| \leq LW_p(\mu, \nu), \quad |g[\mu](x) - g[\nu](x)| \leq LW_p(\mu, \nu);$$

where

$$W_p(\mu, \nu) = \inf \left\{ \int_{\mathbb{R}^{2d}} |x - y|^p d\pi(x, y) \mid \pi \in \Pi(\mu, \nu) \right\}^{1/p},$$

is the Wasserstein distance.

Existence and uniqueness

$$\partial_t \mu + \nabla \cdot ((f[\mu] + ug[\mu])\mu) = 0. \quad (2)$$

Proposition

Assume (H1) – (H2) – (H3). If $(t, x) \mapsto u(t, x)$ is a uniformly Lipschitz in x for every $t \geq 0$ and bounded. Then, given any initial data $\mu(0) = \mu_0 \in \mathcal{P}_c(\overline{B(0, R)})$,

- Equation (2) has a unique solution $\mu(\cdot) \in C^0(\mathbb{R}, \mathcal{P}_c(\overline{B(0, R)}))$.
- If $\mu_0 \in \mathcal{P}_c^{ac}(\overline{B(0, R)})$, then $\mu(t) \in \mathcal{P}_c^{ac}(\overline{B(0, R)})$ for every $t \geq 0$.
- Let Ψ the flow of diffeomorphisms of $\mathbb{R}^d$ generated by the time-dependent vector field $f[\mu] + u(t, \cdot)g[\mu]$, we have

$$\mu(t) = \Psi(t)_\# \mu_0,$$

Where $\Psi(t)_\# \mu_0(A) = \mu_0(\Psi(t)^{-1}(A))$ for every $A \subset \mathbb{R}^d$.

Control constraints - Sparsity

For every $c > 0$ we consider controls of the form

$$u(t, x) = \chi_{\omega(t)}(x)v(t, x)$$

where, for every $t \geq 0$,

- **Sparsity space constraint:** $|\omega(t)| \leq c$,
 - **Finite strength:** $\|u(t, \cdot)\|_{L^\infty} \leq 1$.
-
- $v(t, x)$ is Lipschitz in x for every $t \geq 0$
 - Sparsity space constraint is the infinite dimensional analogue of *decentralized control*

Lyapunov function

Assume that there exists $V : \mathcal{P}_c(\mathbb{R}^d) \rightarrow \mathbb{R}$ such that:

(H4) $t \mapsto V[e^{tf}\mu]$ and $t \mapsto V[e^{tug}\mu]$ are C^1 for all $\mu \in \mathcal{P}_c(\overline{B(0, R)})$ and $u \in \mathbb{R}^d$;

(H5) The uncontrolled system is dissipative, i.e.,

$$\mathcal{L}_f V[\mu] := \lim_{t \rightarrow 0} \frac{V[e^{tf}\mu] - V[\mu]}{t} \leq 0, \quad \text{for all } \mu \in \mathcal{P}_c(\overline{B(0, R)})$$

(H6) Continuity of the Lie derivatives of V up to order 2.

Define

$$\mathcal{Z} = \{\mu \in \mathcal{P}_c(\mathbb{R}^d) \mid \mathcal{L}_f V[\mu] = 0 = \mathcal{L}_{ug} V[\mu] \quad \forall u \in \text{Lip}(\mathbb{R}^d, \mathbb{R})\}.$$

Controllability result

Theorem (C., Piccoli, Rossi, Trélat, 2017)

Assume (H1 – 6). There exist an admissible control $u(t, x) = \chi_{\omega(t)}(x)v(t, x)$ such that the unique solution $\mu(t) \in \mathcal{P}_c^{ac}(\overline{B(0, R)})$, $t \geq 0$ of

$$\partial_t \mu + \nabla \cdot ((f[\mu] + u(t, \cdot)g[\mu])\mu) = 0, \quad \mu(0) = \mu_0,$$

where $\mu_0 \in \mathcal{P}_c^{ac}(\overline{B(0, R)})$,

converges to $\mathcal{Z} \cap \mathcal{P}_c(\overline{B(0, R)})$,

i.e. there exists $\nu(t) \in \mathcal{Z} \cap \mathcal{P}_c(\overline{B(0, R)})$ for each $t \geq 0$ such that $\mu(t) \rightharpoonup \nu(t)$, i.e. for all $\phi \in C_c^\infty(\mathbb{R}^d)$ it holds $\lim_{t \rightarrow \infty} \int \phi d(\mu(t) - \nu(t)) = 0$.